Huawei Access Node IND-ACC-58: INDiranagar Exchange Hardware/Environmental Fault Resolution SOP

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1234

Description

The IND-ACC-58 node failure on the Indiranagar Exchange is attributed to a Hardware/Environmental Fault, specifically the occurrence of ASIC memory parity errors, line card crashes, or power supply redundancy loss on the Huawei Access equipment. This failure domain is a direct result of the intricate interplay between the ASIC (Application-Specific Integrated Circuit) architecture and the environmental conditions at the Indiranagar Exchange site. The ASIC, being the primary component responsible for processing and forwarding traffic on the node, is prone to errors and crashes due to factors such as overheating, power supply fluctuations, or manufacturing defects. Furthermore, the Huawei Access equipment, being a complex system consisting of multiple line cards and power supplies, is susceptible to cascading failures in the event of a primary component failure. The occurrence of ASIC memory parity errors, line card crashes, or power supply redundancy loss triggers an ALARM_1234 with a Major severity level, necessitating immediate intervention to prevent network downtime and data loss.

Pre-requisite Checks

- Pre-requisite: Confirm the presence of remote hands at the Indiranagar Exchange site

- Pre-requisite: Verify the status of redundant links and power supplies on the IND-ACC-58 node

- Pre-requisite: Ensure all necessary safety protocols and procedures are in place before initiating repairs

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display power supply, Expected Output: Power supply 1: Normal, Power supply 2: Normal, Description: This command is used to verify the status of the power supplies on the node, ensuring that both primary and redundant power supplies are functioning correctly.

Command: display board, Expected Output: Board 1: Normal, Board 2: Error, Description: This command is used to verify the status of the line cards on the node, identifying any potential issues or errors on
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the line cards.

Command: display asic, Expected Output: ASIC 1: Normal, ASIC 2: Error, Description: This command is used to verify the status of the ASICs on the node, identifying any potential issues or errors on the ASICs.
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Resolution Steps

Step 1: Step: Replace faulty power supply, Command: replace power supply, Description: This step involves replacing the faulty power supply with a new one, ensuring that both primary and redundant power supplies are functioning correctly.

Step 2: Step: Replace faulty line card, Command: replace board, Description: This step involves replacing the faulty line card with a new one, ensuring that all line cards on the node are functioning correctly.

Step 3: Step: Replace faulty ASIC, Command: replace asic, Description: This step involves replacing the faulty ASIC with a new one, ensuring that all ASICs on the node are functioning correctly.

Post-Check Verification

```
Command: display power supply, Expected Output: Power supply 1: Normal, Power supply 2: Normal, Description: This command is used to verify that the replaced power supply is functioning correctly.

Command: display board, Expected Output: Board 1: Normal, Board 2: Normal, Description: This command is used to verify that the replaced line card is functioning correctly.

Command: display asic, Expected Output: ASIC 1: Normal, ASIC 2: Normal, Description: This command is used to verify that the replaced ASIC is functioning correctly.
```

Roll-back Procedures

- Step: Restore original power supply, Command: undo replace power supply, Description: This step involves restoring the original power supply, ensuring that the node is returned to its original state.

- Step: Restore original line card, Command: undo replace board, Description: This step involves restoring the original line card, ensuring that the node is returned to its original state.

- Step: Restore original ASIC, Command: undo replace asic, Description: This step involves restoring the original ASIC, ensuring that the node is returned to its original state.